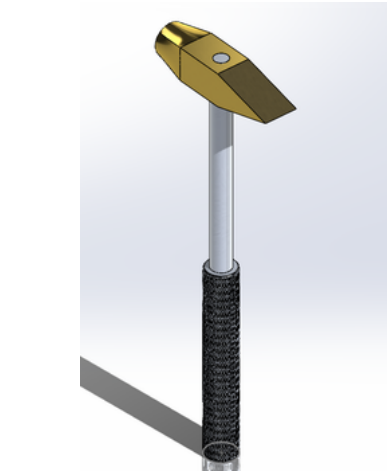
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MECHANICAL HAMMER DESIGN PROJECT







What?

- Designed and fabricated a fully functional hammer from raw stock.

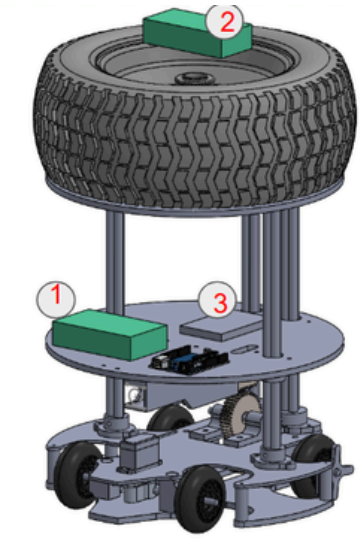
How?

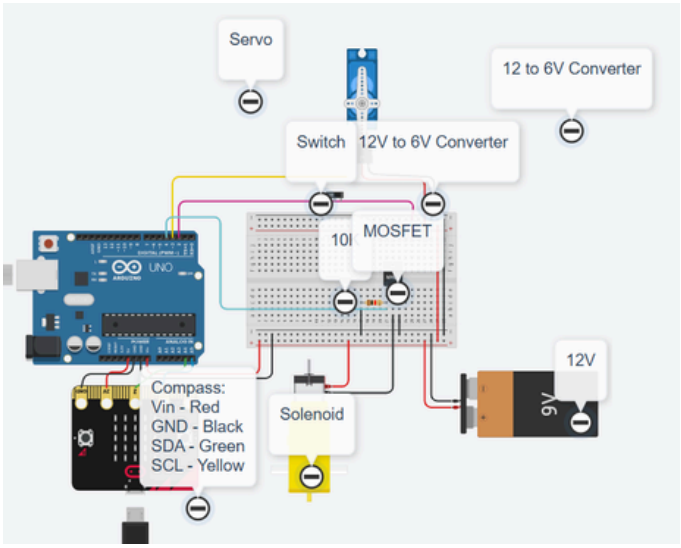
- Designed on **Solidworks**
- Machined components using **manual lathe, vertical mill,** and **horizontal/vertical bandsaw**.
- Performed precision measuring and finishing using **calipers** and **micrometer.**

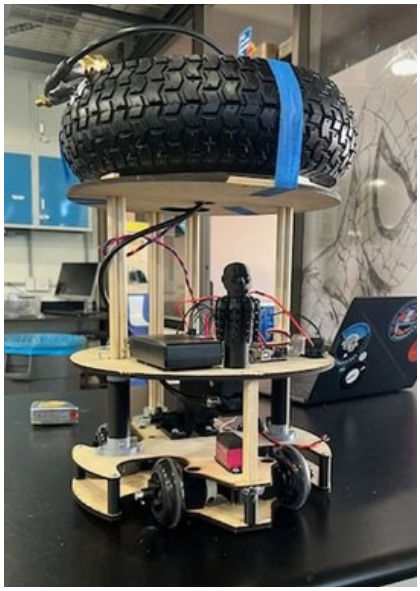
Results

- Produced a functional hammer within ±0.010 in. dimensional tolerance.
- Met all design specifications for fit, finish, and usability.

PNEUMATIC POWERED RACK & PINION ROBOT







What?

- Designed and manufactured an autonomous pneumatically powered robot capable of completing an obstacle course.

How?

- Designed electrical schematics and soldered, wired, and integrated all electrical & pneumatic subsystems.
- Fabricated components using **laser cutting, 3D printing**, and **drill press.**
- Implemented a **PID controller** with **magnetometer** feedback for **autonomous navigation** and heading correction.

Results

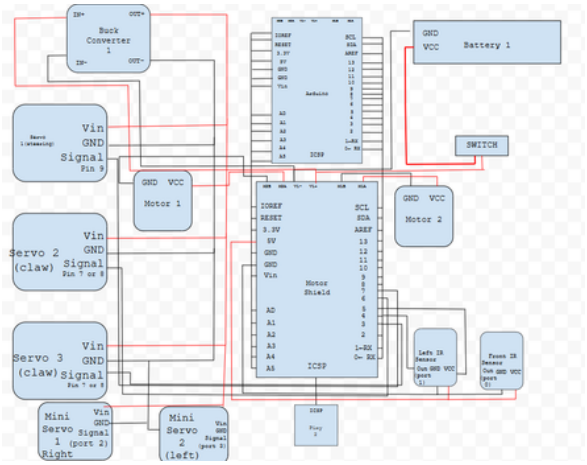
- Competed in a class-wide robotics competition.
- Earned 1st place for overall robot performance.

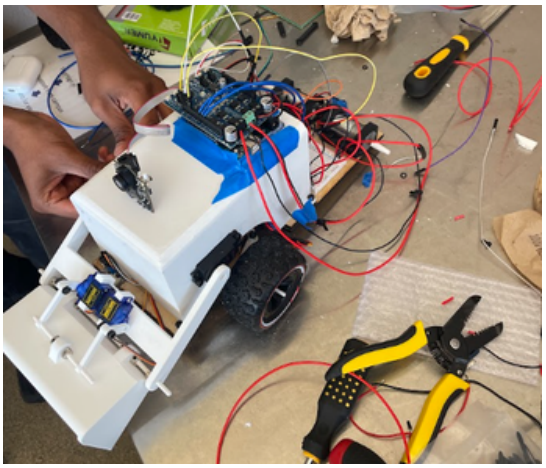
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SPACE EXPLORATION ROVER - SEDS





What?

- Designed and built an autonomous rover that locates, navigates, lifts payloads, and returns to its home base.

How?

- Developed the **electrical system diagram** and sourced all components.
- Procured and integrated **batteries**, **motors**, **cameras**, **IR sensors**, and **servos**.
- Completed all **wiring**, **soldering**, and **electrical assembly**.

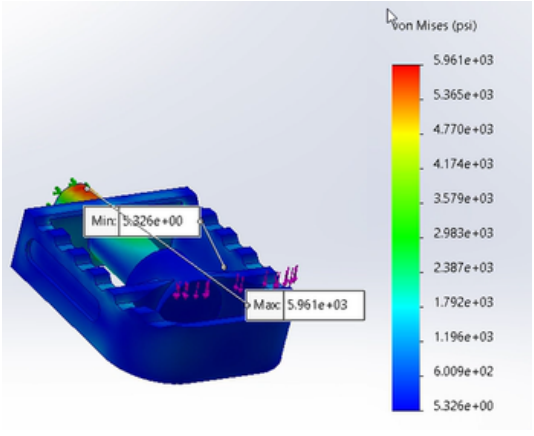
Results

- Robot capable of autonomous navigation to a payload using IR sensors and camera feedback.
- Able to lift and transport payloads using an actuated bucket mechanism.

BICYCLE CAD ASSEMBLY DESIGN







What?

- Created a detailed 3D CAD model of my personal bicycle in SolidWorks.

How?

- Developed the complete **assembly** in **SolidWorks** using **reference geometry**, **sweeps**, and **lofts**,
- Utilized **3D sketches, sweep** features, and **tangent planes** to accurately model the bicycle frame geometry.
- Applied **lofted features** to create the **ergonomic** bicycle seat and other complex surfaces.

Results

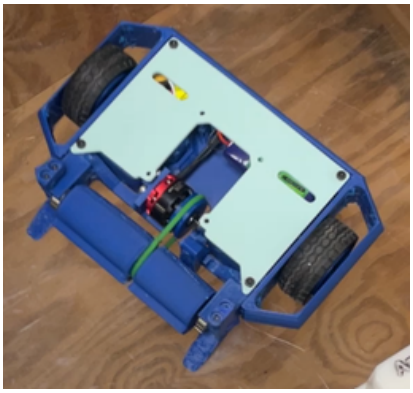
- Completed a fully constrained bicycle assembly consisting of 23 unique parts and 4 subassemblies.
- Performed Finite Element Analysis (FEA) to validate structural integrity under expected operating loads.

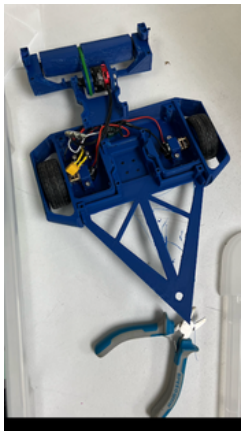
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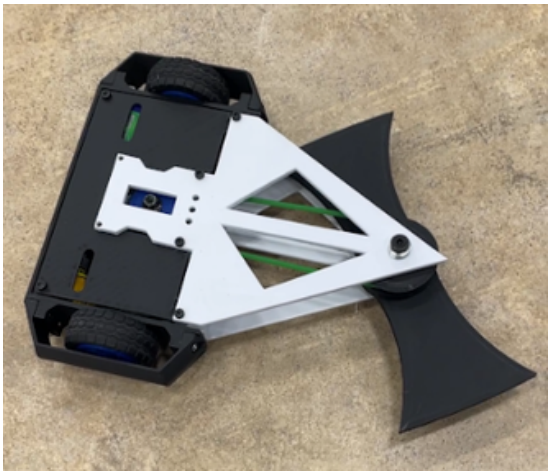
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FEATHERWEIGHT MODULAR BATTLEBOT - ACE ROBOTICS







What?

- Designed and built a 1 lb modular BattleBot with interchangeable weapon systems for competitive robotics.

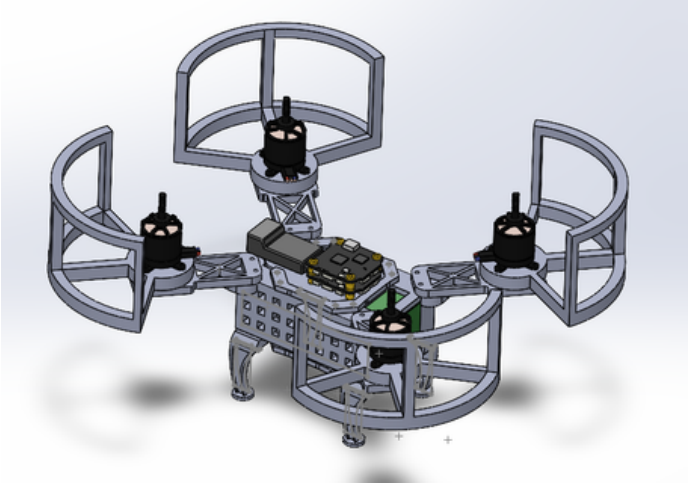
How?

- Contributed to the **CAD** design using **SolidWorks**.
- **Manufactured** structural and custom components using **3D printing**.
- **Routed**, **soldered**, and **integrated** the majority of the robot's electrical system, including **power distribution** and wiring.

Results

- Top 4 Placement at ACE Antweight Assault.
- Successfully used a modular weapon architecture, allowing rapid configuration between horizontal spinner and drum spinner weapon systems

DRONE PROJECT - PETERWORKS





What?

- Designed and manufactured a custom drone for the PeterWorks Drone Competition, optimized for stable flight and lightweight performance.

How?

- Modeled the **airframe** and components in **SolidWorks**.
- **Fabricated** structural components using **3D printing**.
- Selected and integrated the optimal **motors**, **propellers**, **ESCs**, and **battery** to balance **thrust**, **endurance**, and **weight**.

Results

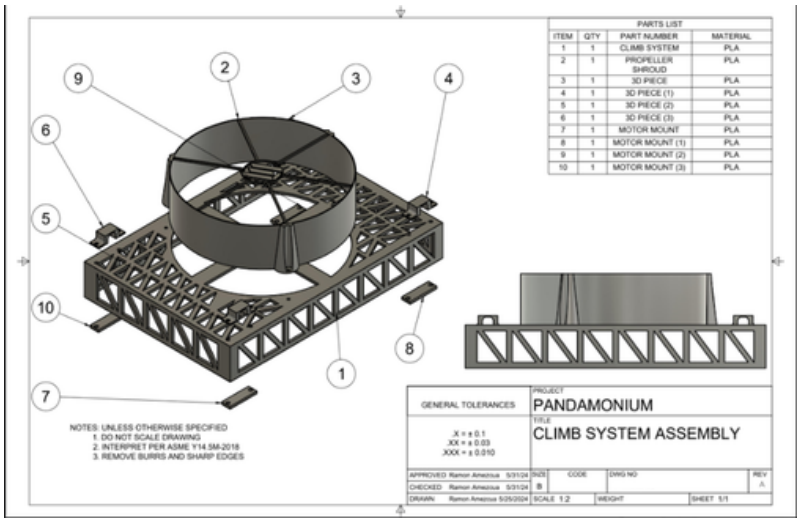
- Successfully competed in the PeterWorks Drone Competition.
- Achieved a successful powered flight.

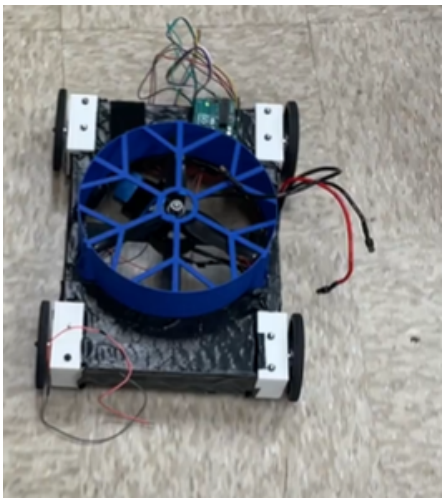
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VERTICAL WALL CLIMBER - MONTERA





What?

- Design and fabricate a device that is able to climb a desired wall for a runtime of 2 minutes.
- Mount and dismount wall easily.

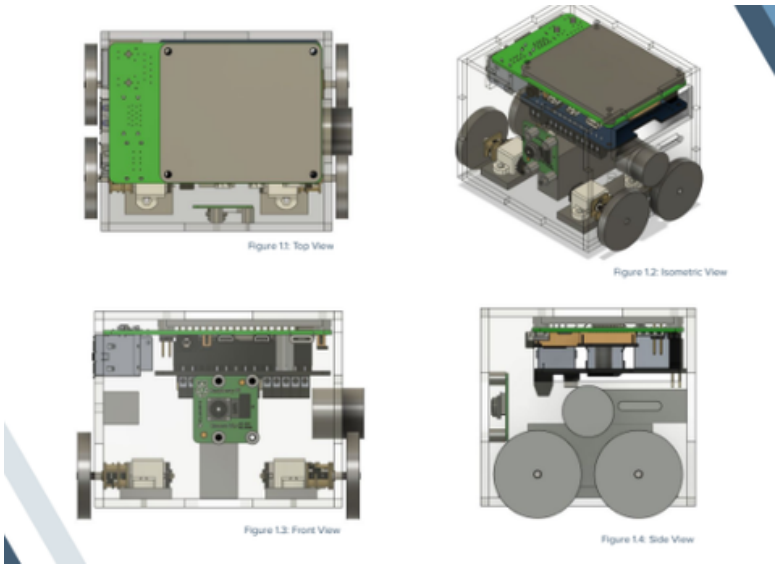
How?

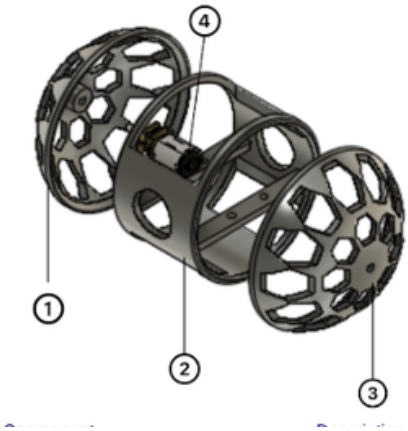
- Helped in **design process** for module. (Made in **Fusion 360)**
- **Soldered** connections to all motors.
- Routed wiring to **Raspberry Pi** and **H-Bridge Stepper.**

Results

- Design was able to stick to wall for a large amount of time.
- Was able to move adequately using remote steering device.

MOVEMENT MODULES - MONTERA





What?

- Module must be able to move and locate sister modules.
- Must be able to attach to sister modules.

How?

- Worked on **Electrical** Aspects
- **Soldered** Connections on all 4 motors.
- Chose **Battery** and **Motor** Components.
- **Paired** Modules to Remote Steering Device.

Results

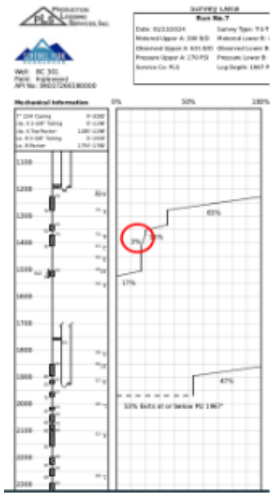
- Modules had the capability to locate each other and move to desired location.
- Able to move all directions across different terrain.

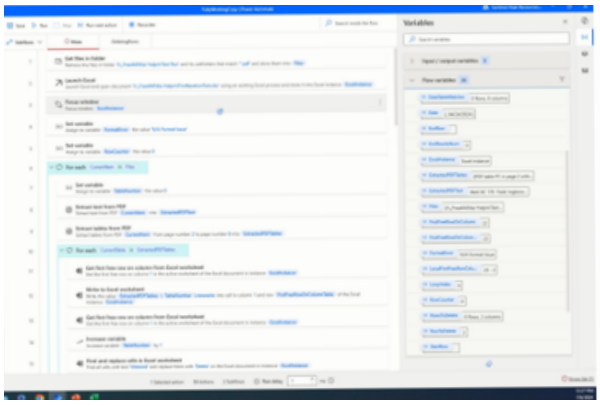
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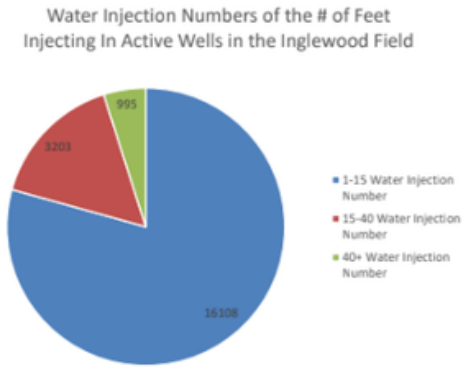
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WELL INJECTION ANALYSIS - SENTINEL PEAK RESOURCES







What?

- Create a database to identify problem wells and injection levels using the catalog of Injection Profile Reports.
- Needed to capture: Historical Data, Geological Intervals, Specific Loss Intervals.

How?

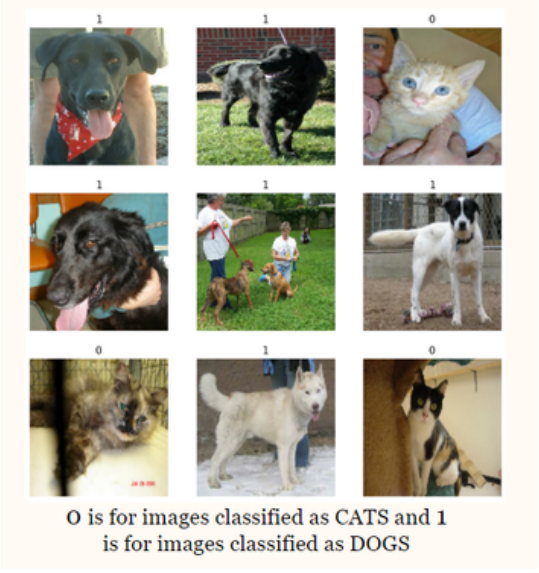
- Used **data collection** features in **Power Automate Desktop** to analyze over 1000+ Injection Profile Reports.
- Applied **Data Integration techniques** through **Excel.**

Results

- Identified over 1000 intervals in 193 wells not injecting adequate amounts of water.
- Found 103 intervals in 62 wells over ejecting.
- Improved Efficiency in Field Wide Production.

IMAGE CLASSIFICATION - TECHNOLOGICAL ADAPTIVE LEARNING INTERNSHIP

```
loss: 0.6542 - accuracy: 0.6387 - val_loss: 0.7178 - val_accuracy: 0.4957
loss: 0.5010 - accuracy: 0.7547 - val_loss: 0.9846 - val_accuracy: 0.4957
loss: 0.3917 - accuracy: 0.8242 - val_loss: 1.2963 - val_accuracy: 0.4957
loss: 0.3189 - accuracy: 0.8636 - val_loss: 0.4946 - val_accuracy: 0.7405
loss: 0.2570 - accuracy: 0.8918 - val_loss: 0.3366 - val_accuracy: 0.8518
loss: 0.2210 - accuracy: 0.9076 - val_loss: 0.3127 - val_accuracy: 0.8723
loss: 0.1990 - accuracy: 0.9173 - val_loss: 0.2557 - val_accuracy: 0.8864
loss: 0.1818 - accuracy: 0.9254 - val_loss: 0.6338 - val_accuracy: 0.7789
loss: 0.1767 - accuracy: 0.9258 - val_loss: 0.1990 - val_accuracy: 0.9203
loss: 0.1519 - accuracy: 0.9390 - val_loss: 0.2171 - val_accuracy: 0.9124
```



0 is for images classified as CATS and 1 is for images classified as DOGS

What?

- Build a machine learning model to differentiate between Cat/Dog Images.

How?

- Used **Python Coding,** my group utilized **Keras/Tensor Flow** to create a convolutional **neural network** that is capable of learning to differentiate between Cat/Dog Images.

Results

- Keras Image Classification Model returned an average model accuracy of 91% after 10 epochs.